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Research article

FEATURES OF AUTOMATIC CONTROL OF TECHNOLOGICAL PARAMETERS OF WATER LEVEL IN THE DRUM STEAM BOILERS

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Abstract

The three-pulse system of automatic control of the water level in the boiler drum received maximum distribution in the energy sector, as well as its modifications, which have the following disadvantages : the presence of static control errors at the end of the transition process in the development of internal disturbances and disturbances of superheated steam flow; large level deviations with sharp and significant changes in the load, which often leads to unloading or shutdown of the boiler or power unit. To eliminate these drawbacks, invariant systems for automatic regulation of the power supply of the drum boiler were previously developed. In work the system of automatic control of technological parameters of water level in a drum of steam boilers is considered. The description of control systems and a detailed description of the electrical power systems of the control system are given. Various methods of tuning the PI controller of a typical three-pulse automatic power control system are considered. The application of the method of structural-parametric optimization and the theory of invariance makes it possible by changing the structure of the system and optimizing the dynamic adjustment of the automatic power control system to significantly improve the quality of maintaining the water level in the boiler drum. Invariant automatic control systems eliminate static control errors under all influences and can significantly improve the quality of maintaining the level under variable conditions. The methods are proposed to significantly improve the quality of water level control in the boiler drum, eliminating the shortcomings of the typical automatic control system. The results of the work can be used in the design of power control systems of steam drum boilers. The results obtained can be used in the development of adaptive control systems for other thermal power facilities.

Keywords: automatic control of power supply, PI regulator, Laplace transforms, transient, regulator coefficients, water level, drum steam boiler.

摘要：鍋爐汽包水位自動控制的三脈衝系統在能源領域得到最大限度的分配及其修改，具有以下缺點：在過渡過程結束時存在靜態控制誤差。過熱蒸汽流的內部擾動和擾動的發展;大的水平偏差，負荷急劇且顯著變化，這通常導致鍋爐或動力裝置的卸載或關閉。為了消除這些缺點，先前開發了用於自動調節鼓式鍋爐的電源的不變系統。在工作中，考慮了蒸汽鍋爐水位技術參數的自動控制系統。給出了控制系統的描述和控制系統的電力系統的詳細描述。考慮了調整典型三脈沖自動功率控制系統的 PI 控制器的各種方法。結構參數優化方法和不變性理論的應用使得改變系統結構和優化自動功率控制系統的動態調整成為可能，從而顯著提高鍋爐水位維持水質。 。不變的自動控制系統消除了所有影響下的靜態控制誤差，並且可以顯著提高在可變條件下維持水平的質量。提出了一種顯著提高鍋爐水位控制質量的方法，消除了典型自動控制系統的缺點。該工作的結果可用於汽包鍋爐的功率控制系統的設計。獲得的結果可用於開發其他火力發電設

施的自適應控制系統。

关键词: 自動控制電源，PI 調節器，拉普拉斯變換，瞬態，調節係數，水位，鼓式蒸汽鍋爐。

I. INTRODUCTION

Automation of power supply of drum boilers provides automatic control of water supply under normal flow of operating modes of the boiler, and in the modes of starting and stopping the boiler unit. In turn, normal operating modes can occur at a constant and variable (sliding) pressure of fresh steam.

As an indicator of compliance of the material balance between steam and water – the flow of fresh steam and feed water serves the level of the boiler drum. The deviation of the water level in the drum from the average value characterizes the presence of unbalance between the flow of feed water and steam flow. It (deviation) also occurs due to changes in the steam content of steam in the steam-water mixture of lifting pipes due to fluctuations in the steam pressure in the boiler drum or changes in the thermal perception of the evaporating heating surfaces.

Thus, with an increase in steam consumption at the first moment after the disturbance, the water level in the drum increases as a result of a sharp decrease in steam pressure, which in turn leads to an increase in the steam content in the lifting pipes of the circulation circuit and an increase in the level. This phenomenon is called level swelling.

When changing the load of the boiler and, as a consequence, changing its steam capacity, the average water level should be maintained constant. The maximum permissible deviation of the water level in the drum is +100 mm from the average value set by the manufacturer. In which case, the average level does not necessarily have to coincide with the geometric axis of the drum. Lowering the level below the visible part of the water-measuring glass installed on the drum of the boiler unit is considered a “loss” of water, and exceeding its upper visible part is referred to as “overfilling”. The distance between these critical marks is 400 mm.

Lowering the level below the connection point of the downpipes of the circulation circuit can lead to power failure and water cooling in the lifting pipes, a violation of their strength in the areas of docking with the drum body, and in the most severe cases, a burnout.

II. METHODS AND MATERIALS

An excessive increase in the level can lead to the deterioration of the action within the drum separation devices, the salt drift of the superheater as well as the water particles being displaced into the turbine, which can cause severe mechanical damage to the blades of its rotor. The supply of water to the drum is carried out through one line – and in some rare cases, two lines – of feed water pipelines, and in the case of the latter, one of them serves as a backup.

The scheme of automatic control of the power supply of the boiler unit is as follows: The principle of combined perturbation control is implemented in the automatic control system (ACS) of the boiler water supply – when changing the flow of steam or feed water and deviation – when changing the water level in the boiler drum.

The power regulator should ensure the constancy of the average water level regardless of the boiler load and disturbance (Figure 1).

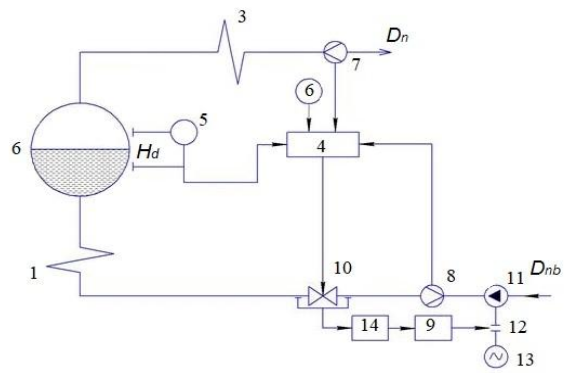


Figure 1. Schematic diagram of boiler drum power control: 1-economizer, 2-boiler drum, 3-superheater, 4-power regulator, 5-level sensor, 6-setting device, 7-steam flow sensor, 8-feed water flow sensor, 9-capacity regulator, 10-feed valve, 11-feed pump, 12- hydraulic coupling, 13-electric motor, 14 – differential pressure gauge.

In the power supply of ACS, a three-pulse power regulator is used for this purpose. Signals in the perturbation: the fresh steam flow D_n, feed water flow rate D_{nb}. Deviation signal: level in the drum of the boiler unit H_d. The signal flow of feed water is used as the switching for removing static signal on the steam flow.

The power regulator moves the adjustment organ on the feed water line when an unbalanced signal appears between the feed water flow and the superheated steam. In addition, it affects the position of the valve when the water level in the drum of the boiler unit deviates from the set value. The use of signals D_n and D_{nb} provide

high-speed ACS power supply H_d – a given accuracy of maintaining the level in the drum.

In the scheme of the measuring unit of the power regulator sensors D_n , D_{nb} and H_d are included in such a way that when the water level in the boiler drum decreases, increasing the flow of steam, and reducing the flow of feed water, they act in one direction – in the direction of opening the feed valve, and with an increase in the level, reducing the flow of steam and increasing the flow of feed water in the direction of closing the feed valve.

Slide valves and spool-type valves are used as regulating power supplies.

When the load on the boiler is completely relieved due to the increase in the steam pressure in the drum, the safety valves may be activated. The amount of steam passing through these valves is not taken into account by the steam flow sensor. The power regulator thus becomes a two-pulse and will maintain an understated level in the drum in accordance with the uneven level of the regulator. Therefore, it is necessary to choose the minimum possible amount of unevenness in terms of level, providing acceptable dynamic quality of ACS power.

With a sliding (variable) initial steam pressure before the turbine for the boiler working with it in the block, there is no throttling of the working fluid throughout the steam-water path, and the pressure is reduced by reducing the rotation speed of the feed pump, this reduces the power required to drive it. The change in the number of revolutions of the feed pump (11) connected by the hydraulic coupling (12) with the electric motor (13) is achieved by the action of the performance regulator (9) by the pressure drop signal on the feed valve (10) from the diphmanometer (14).

The maximum permissible deviations of the water level in the boiler drum of a particular type are set by the manufacturer. Therefore, maintaining a given water level in the boiler drum is an important technological parameter, on which also depends the reliability, safety and efficiency of the boiler equipment of the power plant. This requirement is an important and urgent task in their operation. Steam drum boiler units and three-pulse ACS are widely used in power engineering, as well as their modifications [1].

A three-pulse ACS regulator ensures good testing of jump tasks but does not ensure qualitative testing of internal and extreme external disturbances (the presence of a static error of regulation). This property of the regulator will reduce the efficiency, reliability, and durability of the boiler, making it relevant to modernize the regulator. The dynamics of the water level in the boiler drum when disturbed by the flow of fed water is characterized by a lack of self-leveling, as well as the phenomenon of “level swelling” when disturbed by the flow of superheated steam [2].

In a typical three-pulse ACS water supply of a steam generator (boiler), which is most commonly used in thermal power plants [1-3], a proportional-integral regulator is covered by feedback on the flow rate of fed water. The introduction of this pulse gives the regulator the function of a water flow stabilizer, eliminating the dependence of the level on the fluctuations in the pressure drop on the valve. Therefore, the three-pulse regulator is set so that the same change in the consumption of steam and water causes the same movement of the rod for regulating the filling valve. With this setting, a change in the boiler load leads to a corresponding change in water flow, and the ACS goes into a new state of equilibrium without a significant change in the level. The pulse level is corrective, eliminating the inaccuracies of the settings and inconsistencies that arise in the process of regulation, which is especially important for regulating objects without self-leveling [4].

The simplest static controller (P-controller) has a leading angle equal to $\phi = pi$, which enables it to provide a minimum dynamic error of the adjustable parameter in transient modes for objects without self-leveling. The simplest ideal astatic regulator (I-regulator) has a small lead angle equal to $\phi = pi/2$, which is not suitable for objects without self-leveling, since the closed system becomes neutral or unstable. The advance angle of the real integrating regulator occupies an intermediate position between these two regulators, but the dynamic control error is greater than that of the static one. Therefore, in a three-pulse ACS of the water level in the boiler drum, to increase the stability of the regulation, a PI regulator with a rigid feedback is used. In this case, the ACS consists of two circuits: internal, including a regulator and a rigid feedback device, and external, including the object of regulation and an equivalent regulator, consisting of a rigid feedback device and a regulator [3].

The work of a typical three-pulse power supply ACS and its shortcomings were investigated in [5, 6, 17, 18, 19], where the relevance of its modernization for a significant improvement in the quality of regulation in a wide range of boiler loads was examined. The choice of the structure of the control system, the

law of regulation, and the optimal parameters of the dynamic adjustment of the regulator with main effects has a significant influence on the quality of transients of the water level in the boiler drum. Many engineers attempted to improve the quality of the standard three-pulse ACS power supply. In [7] to improve the operation of the three-pulse power regulator, it is recommended to use a circuit with a differentiator, the output of which is signaled by the material unbalance of steam consumption and feed water (Figure 2).

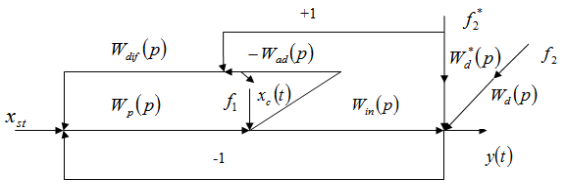


Figure 2. Block diagram of the simulation model three-pulse ACS with the differentiator: $y(t)$ – main adjustable value; x_{st} – setpoint of the main controlled variable; $x_c(t)$ – control action; f_1 – an internal perturbation; f_2 – external fuel outrage; f_2^* – external disturbance of the flow of superheated steam; $W_d(p)$ – the transfer function of the furnace of disturbance; $W_d^*(p)$ – the same external disturbance of the flow of superheated steam; $W_p(p)$ -the same PI-controller; $W_{dif}(p)$ – the same differentiator; $W_{ad}(p)$ – the same advance plot of the object of regulation; $W_{in}(p)$ –the inertial part of the subject regulation.

A BKZ-210-140 steam generator operating on a common steam line – maximum capacity 210 t/h, minimum 140 t/h [3] – was chosen as the object of regulation. The transfer function of the external disturbance by the superheated steam flow rate with the phenomenon of "level swelling" of water in the boiler drum was taken as the difference between the transfer functions of the first-order inertial link and the ideal integrating link [8]

$$W_d^*(p)=W_2(p)-W_3(p)=\frac{k_2}{T_2p+1}-\frac{1}{T_3p}, \quad (1)$$

where $T_2=20$ s; $T_3=67$ s; $k_2=0.75$; and p – Laplace operator.

The transfer function of the flue perturbation took the following form:

$$\frac{k_d}{T_dp+1}=\frac{0.5}{30p+1}, \quad (2)$$

where k_d is the gain of the disturbance, and T_d – the time constant.

The transfer function of the leading section of the feed water flow rate during the abrupt movement of the control feed valve has the form of an inertial link of the first order

$$W_{ad}(p)=\frac{k_{ad}}{T_{ad}p+1}=\frac{1}{5p+1}, \quad (3)$$

where k_{ad} is the transmission coefficient of the leading sector; and T_{ad} – the time constant of the leading sector, characterizing the dynamics of changes in the flow of feed water.

The experimental curve on the water level in the boiler drum under the disturbance of feed water flow can be represented as a real integrating link of the type

$$W_1^*(p)=\frac{1}{T_1p(\tau_1p+1)}, \quad (4)$$

where $T_1=37$ s is the time constant of the real integrating link; and $\tau_1=21$ s is the delay time along the regulatory impact channel [3].

The transfer function of the PI controller is as follows:

$$W_p(p)=\frac{k_p(T_ip+1)}{T_ip}, \quad (5)$$

where k_p – transmission ratio of the controller; T_i – integration time of the controller.

The numerical value of the transmission coefficient of the PI-regulator of the three-pulse ACS is usually calculated by taking into account the transfer function of the object (6) using the formula:

$$k_p=\gamma\frac{T_1}{\tau_1}, \quad (6)$$

where γ is the proportion of the ratio of time constants T_1/τ_1, the numerical value of which is variable for different authors.

For example, in [2] it is recommended to accept $\gamma=1.0$, in [4] – $\gamma=0.8$ at degree of attenuation of transients $\psi=0.75$, in [5] – $\gamma=0.55$ at degree of attenuation of transients $\psi=0.9$.

The transfer function of the external disturbance of the flow of superheated steam with the phenomenon of "level swelling" of water in the boiler drum has the following form [6]:

$$W_d^*(p) = W_2(p) - W_3(p) = \frac{k_3}{T_3 p + 1} - \frac{1}{T_4 p}, \quad (7)$$

i.e. it represents the difference between the transfer functions of the inertial link of the first order and the ideal integrating link. In this case, the greater the value of the "level swelling," the greater the numerical value of the transfer coefficient of the inertial link of the first order k_3.

With the transfer function of the PI-regulator covered by a single feedback, we give the form of the transfer function of the equivalent regulator

$$W_{equ1}(p) = \frac{W_p(p)}{1 + W_p(p)} = \frac{T_i}{T_i\left(\frac{1+k_p}{k_p}\right)p + 1}, \quad (8)$$

i. e. we obtain an integro-differentiation link.

Since $T_i\left(\frac{1+k_p}{k_p}\right) > T_i$ the transfer function of the equivalent regulator (8) in the dynamics represents a link of slow response.

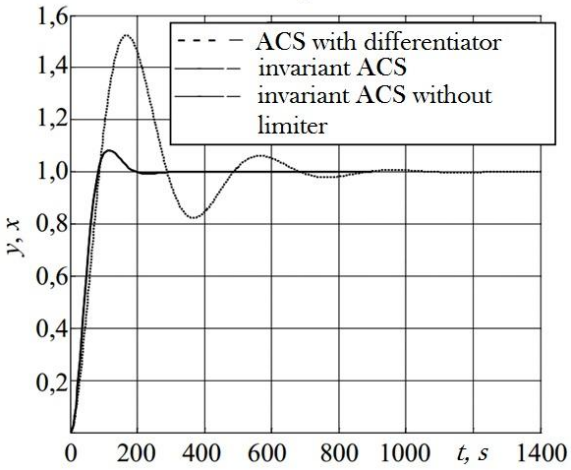
The PI controller was adjusted by the method of symmetric optimum [9]

$$k_p = \frac{T_1}{2k_{ad}\tau_1} = 0.88; \; T_i = 4\tau_1 = 84\,\text{s}. \quad (9)$$

The differentiator parameters were taken according to [10]

$$W_{dif}(p) = \frac{T_1 p}{T_1 p + 1} = \frac{37p}{37p + 1}. \quad (10)$$

a



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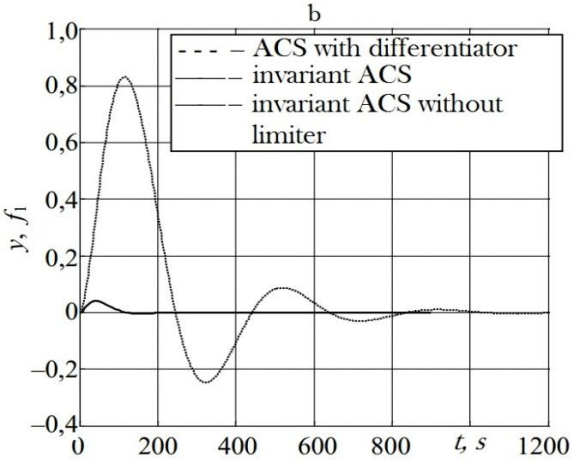

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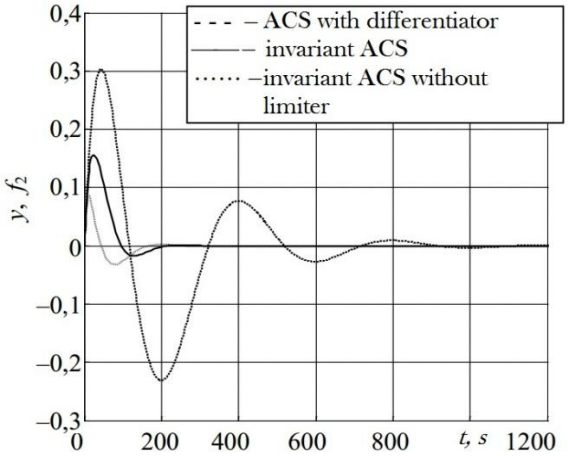

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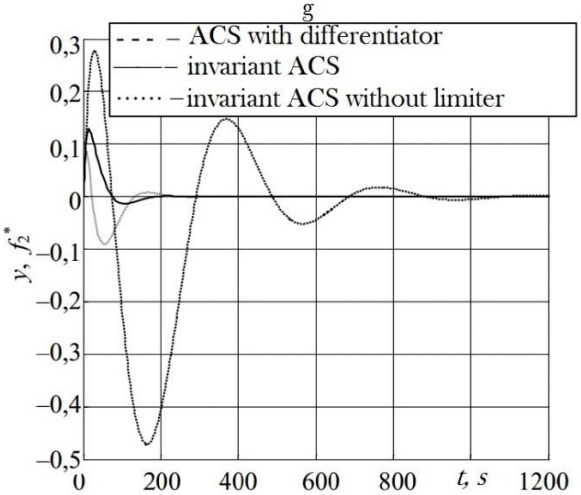

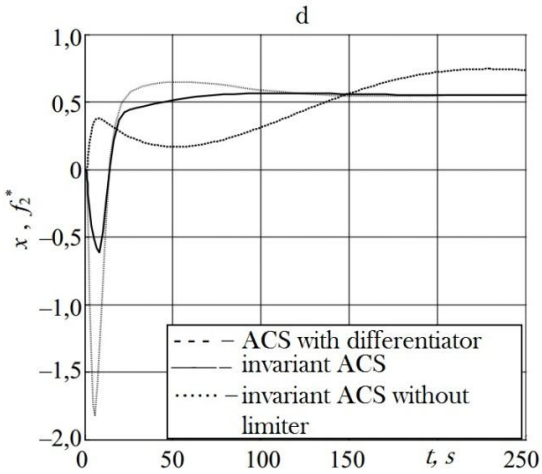

Figure 3. Graphs of transient processes of invariant ACS under the following effects:

a – the practicing a jump of a given impact x_{set}; b – the practicing internal disturbance of f_1; c – the working out external furnace disturbance f_2; g – the working out external disturbances with superheated steam consumption f_2^*; d - the change in the regulatory effect $x_c(t)$ during the development of external disturbances steam consumption; t – time.

Currently, the methods of analytical design of optimal control systems can improve the quality of transients. The results of studies in [5, 6] showed that by changing the structure of the system and optimizing the dynamic setting of the ACS power, the method of structural-parametric optimization and the theory of invariance significantly improved the quality of the water level in the boiler drum. However, this was achieved by increasing the maximum value of the regulatory impact of the feed water flow. At the same time, the value of the latter should be less than the value of the regulatory impact of a typical three-pulse ACS, as it is directly related to the reliability of the boiler equipment, and the durability of the metal drum and water economizer.

To determine the rational structure and optimal parameters of the dynamic adjustment of the proposed ACS (Figure 4), and by providing essential improvement of the quality of maintenance of the water level in the boiler drum at the main influences and considering the maximum size of regulating influence, the transfer function of the optimal regulator [11] is as follows:

$$W_p(p)=\frac{1}{W_{ad}(p)}\cdot\frac{W_{st}(p)}{1-W_{st}(p)}. \qquad (11)$$

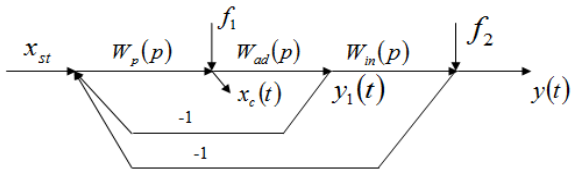

Figure 4. Block diagram of the simulation of dual-circuit ACS: f_2 – external perturbation; $y_1(t)$ – intermediate control value (the rest of the notation is indicated in Figure 2)

In this case, according to the specified effect (optimality criterion), the optimal transfer function of the internal circuit of the system corresponds to the inertial link of the first order:

$$W_{st}(p)=\frac{1}{T_{st1}p+1}, \qquad (12)$$

where T_{st1} is the given time constant of the optimality criterion by the intermediate controlled value.

Substituting the transfer functions (6) and (12) in (11), we obtain the optimal transfer function of the regulator (PI-regulator) with one dynamic setting parameter T_{st1}

$$W_p(p)=\frac{T_{ad}^*p+1}{k_{ad}T_{st}p}=\frac{5p+1}{T_{st1}p}. \qquad (13)$$

When we replace the internal circuit (Figure 4) with the inertial link of the first order (12), we obtain a single-circuit ACS (Figure 5). We write the transfer function of a closed ACS (Figure 5) on the setting influence as:

$$W_{y,x_{st}}(p)=\frac{W_{st}(p)\cdot W_{in}(p)}{1+W_{st}(p)\cdot W_{in}(p)}; \qquad (14)$$

$$W_{y,x_{st}}(p)=\frac{1}{T_1\tau_1T_{st}p^3+T_1(T_{st}+\tau_1)p^2+T_1p+1}. \qquad (15)$$

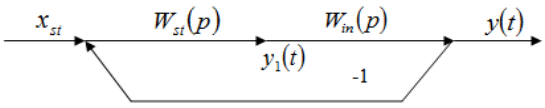

Figure 5. ACS with the replacement of the inner circuit

As a result, the characteristic equation of the third order is obtained by the denominator of the transfer function (15). According to [12],

$$a_0=T_1\tau_1T_{st};\ a_1=T_1(T_{st}+\tau_1);\ a_2=T_1;$$
$$a_3=1; \qquad (16)$$

$$a_0p^3+a_1p^2+a_2p+a_3=0; \qquad (17)$$

$$s^3 + A_2 s^2 + A_1 s + 1 = 0; \quad (18)$$

$$A_1 = \frac{a_2}{\sqrt[3]{a_0 a_3^2}}; \; A_2 = \frac{a_1}{\sqrt[3]{a_0^2 a_3}}, \quad (19)$$

where, A_1, A_2 – the coefficients of Vyshnegradsky, with transients $A_1 = A_2 = 3$ in the system are aperiodic with maximum speed:

$$a_2 = 3\sqrt[3]{a_0 a_3^2}; \; a_3^2 = 27 a_0 a_3^2;$$

$$T_1^3 = 27 T_1 \tau_1 T_{st}. \quad (20)$$

Therefore, considering (7), the optimal numerical value of the time constant T_{st} of the optimality criterion is:

$$T_{st} = \frac{T_1^2}{27\tau_1} = 2{,}41\,\text{s}. \quad (21)$$

To significantly improve the quality of the processing of the external disturbances of the double-circuit ACS (Figure 4), we convert to invariant ACS, the block diagram that is shown in Figure 6. Similar to the real inertial part of the controlled object, we form a dynamic model of this part, the output of which is algebraically added to the basic controlled value. We supply the obtained difference to the input of the equivalent external perturbation compensation device without its measurement, and the output with the reverse sign through the limiter to the input of the regulator.

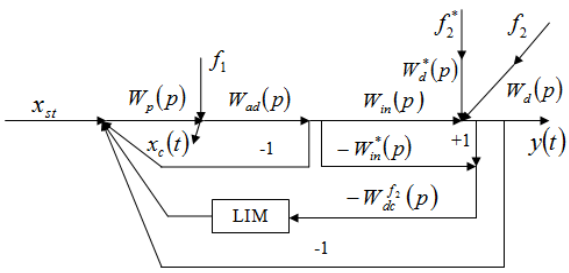

Figure 6. Block diagram of simulation of transient processes invariant ACS supply: $W_{dc}^{f_2}(p)$ – transfer function of a device for the compensation of external disturbances; $W_{in}^*(p) = W_{in}(p)$ – the same models of the inertial part of the regulation object; LIM – the limiter of the correction value of the task (the rest of the notation is indicated in Figure 2).

III. Results and Discussion

We determine the transfer function of the equivalent external perturbation compensation device from the following invariance condition of the main controlled variable from the change of external perturbations:

$$W_{dc}^{f_2}(p) \cdot W_{st2}(p) \cdot W_{in}(p) = 1, \quad (22)$$

From (22) we find the transfer function of the external perturbation compensation device, which for its physical realization, is multiplied by the transfer function of the inertial link of the third order with the same time constants T_{st2}:

$$W_{dc}^{f_2}(p) = \frac{T_1 p(\tau_1 p + 1) \cdot (T_{st} p + 1)}{(T_{st2} p + 1)^3}. \quad (23)$$

Having accepted

$$T_{st2} = T_{st1}, \quad (24)$$

we get

$$W_{dc}^{f_2}(p) = \frac{T_1 p(\tau_1 p + 1)}{(T_{st2} p + 1)^2} = \frac{T_1 p}{T_{st2} p + 1} \cdot \frac{\tau_1 p + 1}{T_{st2} p + 1}. \quad (25)$$

Hence, we have a consistent connection of the differentiating link and the rapid response link, since in the numerical value of τ_1, the numerator is greater than T_{st2} the denominator. Since the difference between the signals of the main controlled value and the output of the model of the inertial section of the controlled object comes before the input of the rapid response link, then to the input of the differentiator, the large signal from the output of the differentiating link should be limited in magnitude to reduce the value of the correction of the regulator task. This will lead to a corresponding reduction in the maximum value of the regulatory impact of the feed water flow.

The diagrams of transient processes show that the perturbing influences of time invariant regulation in the ACS are reduced to five times less, compared to the ACS with the differentiator. When processing the jump in jobs, the magnitude of the overshoot is reduced 4.3-fold, and the maximum dynamic error of regulation—in developing internal perturbations—is reduced by 16.4 times; when developing of the combustion disturbance, it is reduced by 2 times; and when developing the perturbation of the flow rate of steam, it is reduced by 2.1 times. The value of the regulatory impact of the ACS with a differentiator ranges from 0 to +0.75, and the invariant ACS ranges from -0.6 to +0.56.

Limiting the output signal of the external perturbation compensation device leads to a decrease in the maximum value of the control action (Figure 3d). Moreover, the time of processing at external disturbances remains the same, while the maximum dynamic control error at external disturbances somewhat decreases. At the same time, a significant improvement in the quality of regulation was obtained, compared with the standard ACS supply with a differentiator.

IV. CONCLUSION

1. A method has been developed for calculating the economic efficiency of an invariant system of automatic control of the water level in a boiler drum, which considers not only sudden but gradual failures, as well as quality indicators for regulating the water level of a typical (basic) and invariant (new) automated control system under the main influences.
2. The results of the simulation of transient processes showed that when calculating the numerical value of the coefficient γ , the typical three-pulse automatic control system ensures the development of a task jump without a static control error. Moreover, the optimal transients provide the calculation of the numerical values of the coefficient of transfer of the regulator when $\gamma = 2.618$, which is more than recommended in [2, 4, 5]. However, static regulation errors persist during the development of internal and external disturbances with the phenomenon of "level swelling".
3. An invariant system of automatic control of the supply of a drum steam boiler is proposed, with the following distinguishing characteristics:

- the proportional-integral controller of the system has one parameter of dynamic setting, the numerical value of which is chosen from the condition of equality of the Vyshnegradsky coefficients $A_1 = A_2 = 3$. This allows the development of the driver action with maximum speed;
- parallel to the real inertial section of the object of regulation, its dynamic model is formed, which allows the isolation of an equivalent external disturbance without measuring it.

4. Comparing the graphs of transient processes of the automatic power control system with the differentiator and the developed invariant system of automatic power control showed the significant advantage of the latter.
5. Introduction of an output limiter from an external disturbance – compensation device reduces the maximum value of the regulating influence on feed water flow, which is less than that of a typical three-pulse automatic control system, without degrading the quality of transients during the task jump and internal disturbance.

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